

DE Lab5
Hayden Fuller

Q1:

Design and characterize propagation delays for a CMOS inverter using the textbook 130nm process parameters with the following VS points. Adjust PMOS width as needed to achieve the required VS. Use $R_{ON}(NMOS) = 12.5k$, and $R_{ON}(PMOS) = 30k$, when $W/L = 1$.

a. $V_S = 0.6 V_{DD}$

$$V_M = 0.6 * 1.2 = 0.72$$

$$NMOS \text{ overdrive } V_M - V_{TN} = 0.72 - 0.4 = 0.32$$

$$PMOS \text{ overdrive } V_{DD} - V_M - |V_{TP}| = 1.2 - 0.72 - 0.4 = 0.08$$

$$\sqrt{\beta_p / \beta_n} = (0.32 / 0.08)^2 = 16$$

$$W_P / 70 / 270 = 16 ; W_P = 61.7$$

$$W_P = 61.7 W_N$$

b. $V_S = 0.4 V_{DD}$

$$V_M = 0.48$$

$$NMOS \text{ overdrive } 0.48 - 0.4 = 0.08$$

$$PMOS \text{ overdrive } 1.2 - 0.48 - 0.4 = 0.32$$

$$\sqrt{\beta_p / \beta_n} = (0.08 / 0.32)^2 = 0.0625$$

$$W_P / 70 / 270 = 0.0625 ; W_P = 0.24$$

$$W_P = 0.24 W_N$$

c. Compute and compare τ_{PHL} and τ_{PLH} between part (a) and part (b) with a 1pF load capacitor, using the given R_{ON} values above modified by your calculated W_P .

$$\tau = 0.69 R_{on} C$$

$$R_{onn} = 12.5k ; R_{onp} = 30k ; \text{ scales as } 1/W$$

a) $W_P = 62$

$$NMOS \text{ fall time } R_n = 12.5k ; \tau_{PHL} = 0.69 * 12.5k * 1p = 8.6ns$$

$$PMOS \text{ rise time } R_p = 30k / 62 = 484 ; \tau_{PLH} = 0.69 * 484 * 1p = 0.33ns$$

b) $W_P = 0.24$

NMOS same

$$\tau_{PHL} = 8.6ns$$

$$PMOS R_P = 30k / 0.24 = 125k$$

$$\tau_{PLH} = 0.69 * 125k * 1p = 86ns$$

d. Compute and compare τ_{PHL} and τ_{PLH} between part (a) and part (b) with no load capacitor, where C of a $W=130nm$, $L=130nm$ MOSFET = 1fF, using the given R_{ON} values above modified by your calculated W_P .

$$W=1, L=1 \Rightarrow 1fF$$

$$\text{output node sees drain of both transistors, } C = (W_N + W_P) * 1f$$

$$a) C = (1 + 62) * 1f = 63f$$

$$NMOS \text{ fall time } \tau_{PHL} = 0.69 * 12.5k * 63f = 0.54ns$$

$$PMOS \text{ rise time } \tau_{PLH} = 0.69 * 484 * 63f = 0.021ns$$

$$b) C = (1 + 0.24) * 1f = 1.24f$$

$$\text{fall } \tau_{PHL} = 0.69 * 12.5k * 1.24f = 0.011ns$$

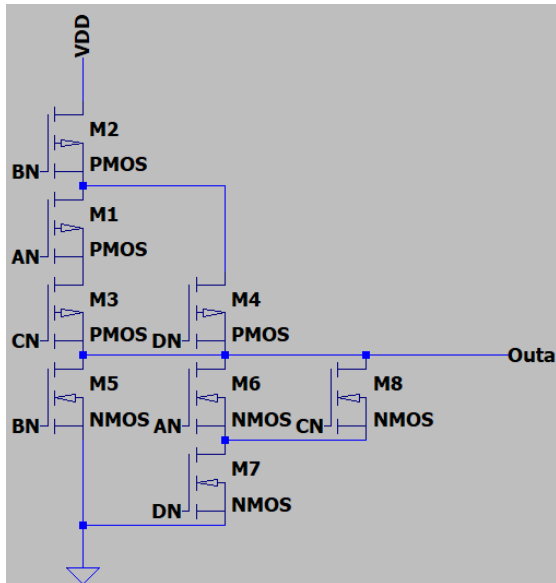
$$\text{rise } \tau_{PLH} = 0.69 * 125k * 1.24f = 0.11ns$$

Q2:

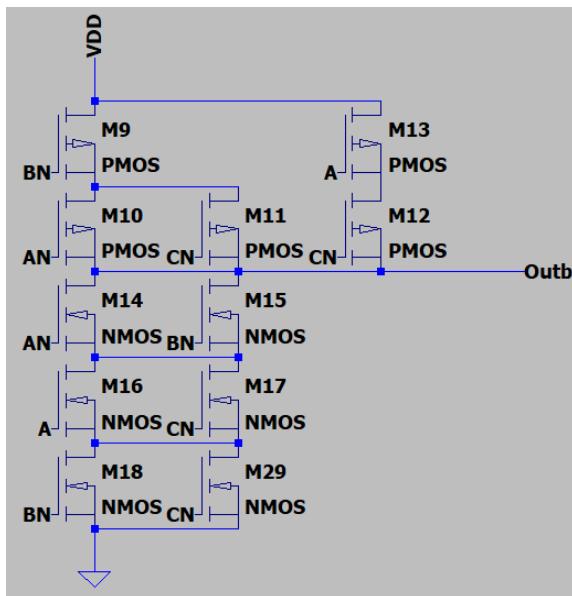
Design static CMOS gates that have the following outputs. You may assume you have the availability of true complement forms of each signal for each input (Ex: Both A and A' are available without an additional inverter). Design each function as a single gate, where one part of your CMOS circuit does not serve as an input for any other section. Device sizing is not required for this problem.

HINT: Remember that NMOS turn on when they receive a high voltage input. Derive the pull-down circuit equation Out' for positive inputs. PMOS turn on when they receive low voltage input. Derive the pull-up equation Out for all inverted inputs.

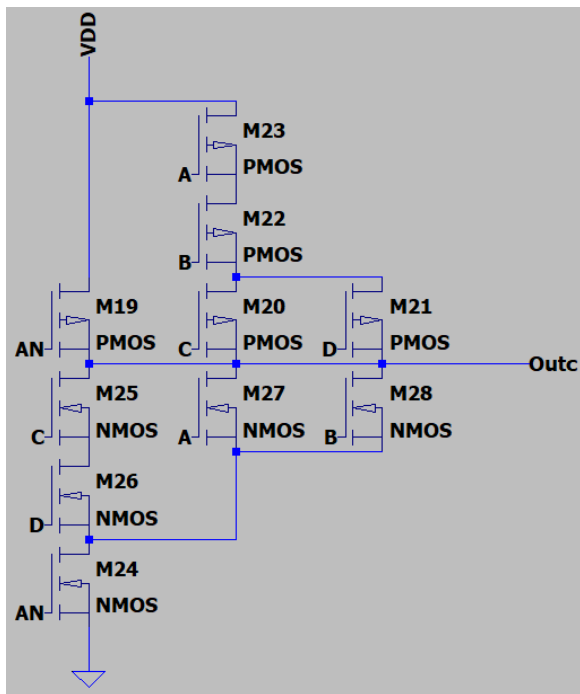
a. $Out = ABC + BD$



b. $Out = AB + (A)C + BC$

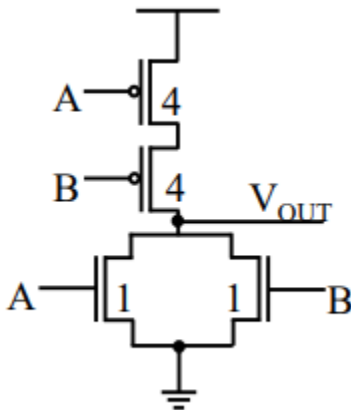


c. $Out = (A + B + CD) + A$



Q3:

Consider a 2-input NOR gate where $W_P = 4$ and $W_N = 1$.



a. Calculate V_S when only input A is switching.

$\beta \propto W/L$

$B=0$, NMOSB off, PMOSB on

$\beta_{\text{PMOS}} = 270 \times 1 = 270$

$\beta_{\text{NMOS (single)}} = 70 \times 4 = 280$

two in series, resistance doubles, β halves, $\beta_{\text{NMOS}} = 140$

$I_{\text{PMOS}} \propto \beta_{\text{PMOS}}(V_M - V_{TN})^2 = 270(V_M - 0.4)^2$

$I_{\text{NMOS}} \propto \beta_{\text{NMOS}}(V_{DD} - V_M - |V_{TP}|)^2 = 140(0.8 - V_M)^2$

$I_{\text{PMOS}} = I_{\text{NMOS}} ; 270(V_M - 0.4)^2 = 140(0.8 - V_M)^2$

$V_S = V_M = 0.567V$

b. Calculate V_S when only input B is switching.

i. NOTE: V_{BS} will affect V_{TP} .

$A=0$, NMOSA off, PMOSA on

node between pmos a and b not at V_{DD}

$|V_{TP}| = |V_{TP0}| + \gamma(\sqrt{2|\phi_F| + V_{SB}}) - \sqrt{2|\phi_F|} = 0.4 + 0.2(\sqrt{0.88 + V_{SB}}) - 0.938$

$V_S \approx (V_{DD} + V_{out})/2 = (V_{DD} + V_S)/2 \approx (1.2 - 0.57)/2 = 0.885$

$V_{SB} = V_S - V_B \approx 1.2 - 0.885 = 0.315$

$|V_{TP}| \approx 0.4 + 0.2(\sqrt{0.88 + 0.315}) - 0.938 = 0.431$

lower still 0.4, upper ≈ 0.431

$270(V_M - 0.4)^2 = 140(1.2 - V_{TP} - V_M)^2 ; V_M = V_S \approx 0.56V$

c. Calculate V_S when both inputs are tied and switching simultaneously.

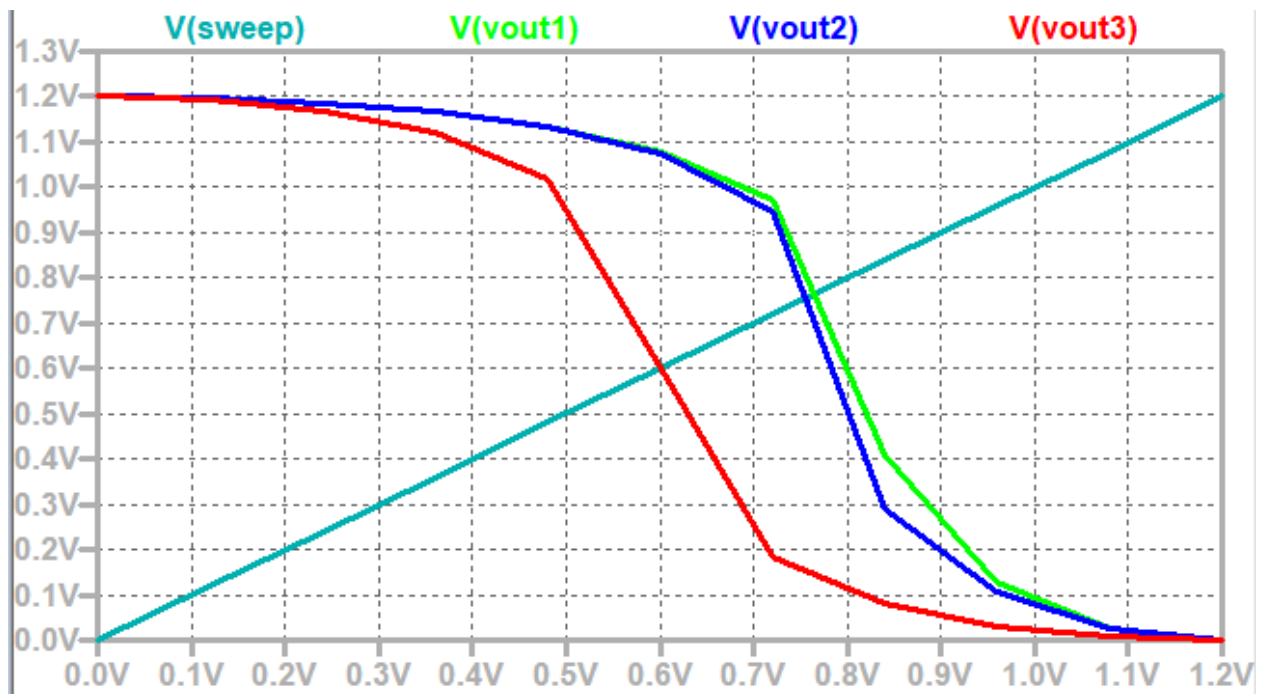
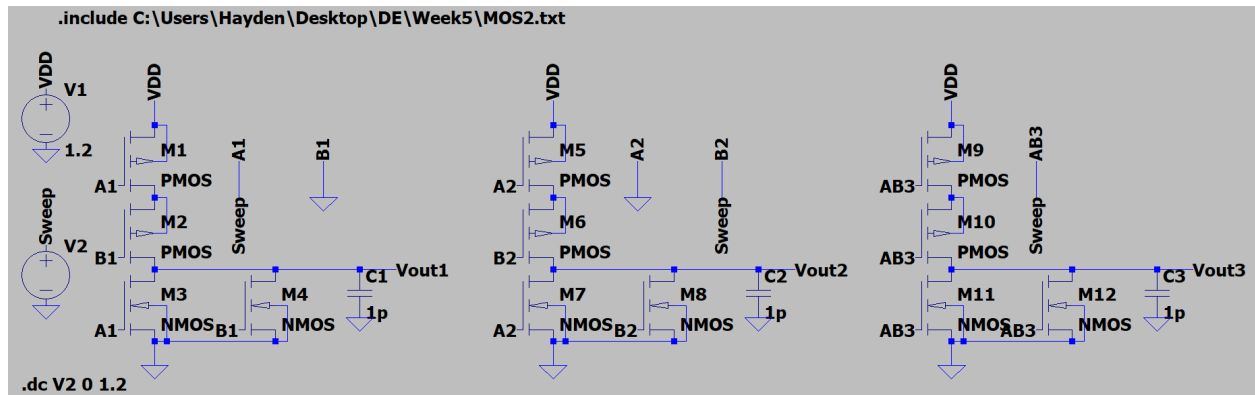
$\beta_{\text{NMOS parallel}} = 2 \times 270 = 540$

$\beta_{\text{PMOS series}} = 140$

$540(V_M - 0.4)^2 = 140(0.8 - V_M)^2$

$V_S = 0.53V$

d. Plot all three VTCs in SPICE.



A switching: 0.764V

B switching: 0.755V

Both tied: 0.6V

e. Compare your calculated vs simulated results.

In my simulation, the NOR gate switches at a much higher point, but the relative values, notably both tied being much lower, are somewhat similar.